

Computer Vision Techniques for Detecting Heterogeneous Robots and Game Elements in Competitive Robotics

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Abstract & Introduction

Autonomous decision-making in VEXU robotics requires reliable real-time detection of both game elements and opposing robots; however, the high variability of robot designs and inconsistent lighting conditions make this a challenging computer vision problem. This work presents a pipeline that integrates color segmentation, classical edge-based object localization, and convolutional neural network (CNN)-based detection to improve robustness in complex field environments. This approach first employs HSV color masking to identify red and blue game pieces, followed by contour analysis to generate bounding boxes. To address failure cases caused by lighting variation and connected contours, an edge-detection module is incorporated that performs Canny filtering, geometric clustering, and circular approximations to isolate both spherical game elements and the structural edges of opposing robots. Finally, a MobileNetV2-based regression CNN and a YOLO model using synthetically generated datasets to detect game elements under diverse background conditions is implemented. Experimental results demonstrate that each method contributes unique strengths: color detection provides high precision under controlled lighting, edge clustering reliably identifies robot structures independent of design variation, and CNN-based models generalize well across backgrounds and occlusions. The primary contributions include (1) a novel hybrid vision framework that unifies classical and deep-learning methods for VEXU field perception, (2) a synthetic-data generation pipeline for efficient CNN training which is also expandable to future seasons with different field elements, and (3) a comparative evaluation of detection strategies under real-world competition constraints. This integrated system offers a foundation for more adaptive autonomous navigation and strategic decision-making in competitive robotics.

Related Work & Background

Real-Time Vision on a Mobile Robot Platform

Autonomous mobile robots in competitive settings such as RoboCup and VEX increasingly rely on on-board computer vision for perception, localization, and decision-making. Early systems in robot soccer employed hand-engineered pipelines that combined color segmentation, blob detection, and simple geometric reasoning to detect balls, goals, and field

lines in real time under tight computational constraints. Sridharan and Stone developed one such vision system, emphasizing efficient segmentation and object recognition on resource-limited robots with rapidly moving cameras. Subsequent work extended these ideas to handle varying illumination through structure-based color learning, demonstrating that robust color-based segmentation remains feasible but requires careful calibration and scene modeling. These approaches motivate the color-thresholding and contour-based methods used as baselines in the present system.

Fast Object Detection By Regression in Robot Soccer

Within competitive robotics, a substantial body of work focuses on vision for robot soccer, where robots must detect balls, teammates, and opponents on constrained fields. Brandão et al. proposed a fast object-detection framework based on regression tailored to robot soccer, highlighting the importance of low-latency detection for real-time decision-making. Lu et al. introduced a robust omnidirectional vision sensor for RoboCup Middle Size League robots, which integrates specialized optics and tailored image processing to obtain reliable field and object perception from a single camera.

You Only Look Once: Unified, Real-Time Object Detection

Modern deep-learning-based detectors such as YOLO have transformed real-time object detection by casting detection as a single-stage regression problem from images directly to bounding boxes and class probabilities. Redmon et al. showed that YOLO can operate at real-time frame rates while maintaining competitive accuracy, making it attractive for embedded robotic platforms. For deployment on resource-constrained hardware, lightweight backbones like MobileNetV2 use depthwise separable convolutions and inverted residual blocks to achieve favorable accuracy–latency trade-offs, and have been successfully integrated into object-detection frameworks such as SSDLite.

Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World

Deep learning-based vision models often require large labeled datasets that are difficult or impractical to collect in robotics competitions such as VEXU. Tobin et al. introduced the concept of domain randomization, demonstrating that models trained entirely on heavily varied synthetic images can generalize effectively to real-world environments without requiring real photographs. Their work shows that object detectors can be made robust by exposing them to randomized textures, lighting conditions, and background clutter during training. This approach is directly relevant to the synthetic-data generation pipeline in the present project, where hundreds of randomized training images of game elements were generated to improve model generalization. By relying on domain randomization principles, the project’s MobileNetV2 and YOLO models achieve improved performance in uncontrolled field environments where lighting and backgrounds differ significantly from training conditions.

A Shared Dataset for Object Detection in Robot Soccer

Developing high-performance object detectors for competitive robotics requires benchmarking across diverse scenes, object orientations, and robot configurations. Pham et al. addressed this need by releasing a standardized, labeled dataset for object detection in robot soccer, containing images of balls, robots, and relevant field structures. Their dataset enables consistent evaluation across different detection architectures and highlights the strengths and weaknesses of CNN-based approaches in dynamic, cluttered environments. This work relates closely to the goals of your project, as the lack of labeled VEX-specific images presents a similar challenge. The project instead constructs its own dataset using synthetic image generation, but follows the same motivation: enabling reliable comparisons between baseline color methods, classical edge detection, and modern CNN detectors. Like Pham et al.'s soccer dataset, the project aims to support accurate detection of both game elements and varied robot designs under competition constraints.

Methodology

Approach Overview

The objective of this project is to develop a unified perception system capable of detecting VEXU game elements—including red and blue balls—and identifying opposing robots with highly variable shapes and geometries. To address this challenge, a hybrid computer vision framework was created that integrates three complementary approaches: (1) HSV-based color segmentation for rapid identification of game elements, (2) edge-based geometric clustering for robot structure detection, and (3) convolutional neural network (CNN)-based object detection trained using synthetically generated data. These components were evaluated independently and in combination to assess their strengths under different environmental conditions.

Technical Details

The baseline detection method employs HSV color masking to isolate red and blue game elements. Frames are converted from RGB to HSV to reduce sensitivity to illumination. Tuned threshold ranges generate binary masks, which are refined through morphological operations. Contours are extracted and filtered based on area and shape, allowing bounding boxes to be drawn around detected game elements. This method is computationally efficient but susceptible to lighting variation and to merged contours when elements overlap or resemble background colors.

Edge Detection and Geometric Clustering

To detect robots with unknown appearance variability, an edge-based detection module was implemented. Gaussian smoothing and Canny edge detection extract structural boundaries within a frame. Contours are grouped into clusters, and geometric properties are used to differentiate spherical game elements from robot structures. Circularity metrics and minimum-enclosing-circle approximations identify balls, while elongated or multi-line clusters

indicate robot edges. This method effectively identifies robots but can also capture unwanted structural edges from field walls or competition barriers.

Convolutional Neural Network Architecture

A lightweight CNN-based detector provides improved robustness to background and lighting variability. Two architectures were tested: a MobileNetV2 regression model that outputs bounding box coordinates, and a YOLO-style single-stage detector that outputs bounding boxes and class probabilities. MobileNetV2's inverted residual blocks and depthwise separable convolutions enable efficient inference. YOLO's unified detection framework allows real-time object recognition with competitive accuracy. Both models were trained with randomized augmentations, including flips, rotations, and brightness shifts, to enhance generalization.

Implementation

OpenCV in Python was used for all classical computer vision components. HSV thresholds and contour heuristics were tuned experimentally. Deep-learning models were implemented using TensorFlow and trained on GPU hardware. The integrated pipeline runs color segmentation and edge detection in parallel threads, with CNN inference executed at a reduced frequency to maintain real-time performance on limited hardware. This modular design balances speed and robustness by combining fast classical detection with more reliable learned detection.

Experimental Design

Performance was evaluated across three controlled test conditions: (1) uniform lighting and simple backgrounds, (2) variable lighting resembling competition environments, and (3) cluttered scenes with overlapping objects and field barriers. Game elements were placed at predefined positions, distances, and occlusion levels. Robot detection was evaluated using robots of varying geometry, reflectivity, and structure. Each configuration was recorded and analyzed across multiple trials to ensure consistency and repeatability.

Data - Synthetic Dataset Generation

Because labeled VEXU data is scarce, a synthetic dataset was created. Cropped images of red and blue balls were superimposed onto randomized backgrounds under varying lighting, scale, and occlusion levels. Between one and five game elements were placed in each frame, and textures were randomized to mimic real competition environments. See **Figure 1**.

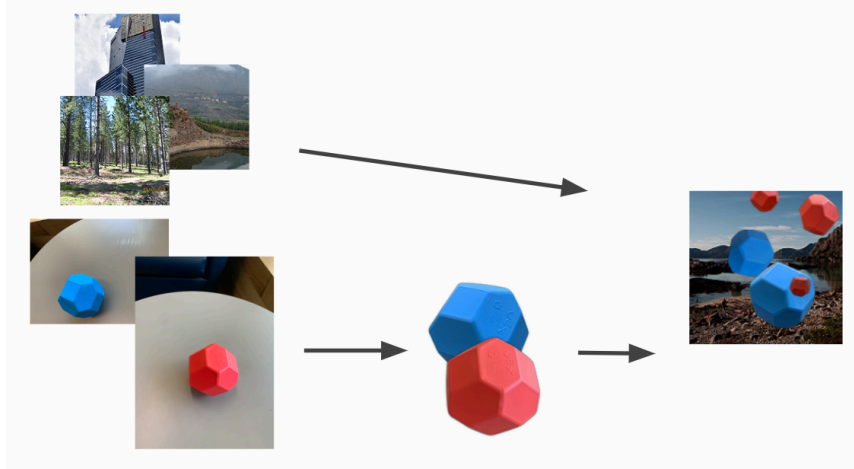


Figure 1: Data generation pipeline

Preprocessing and Annotation

Images were resized to 150×150 pixels. Bounding boxes were stored in Pascal VOC XML format. Augmentations—including brightness variation, edge blurring, and contrast adjustment—were applied to increase dataset diversity.

Ethical and Practical Considerations

Synthetic data eliminates privacy concerns and ensures no human subjects are involved. Care was taken to generate images representative of real VEX competition conditions to avoid overfitting to artificial textures or backgrounds.

Evaluation Metrics

There are no clear numeric metrics used for evaluation. Primarily pipeline is evaluated by human observation on its bounding box drawings on various images.

Experiments & Results

The evaluation for this pipeline was primarily done by observing the various outputs. As seen in **Figure 2** the color detection model was able to clearly pick up every element but also has some extra noise with parts of the image that are similar colors of the elements.

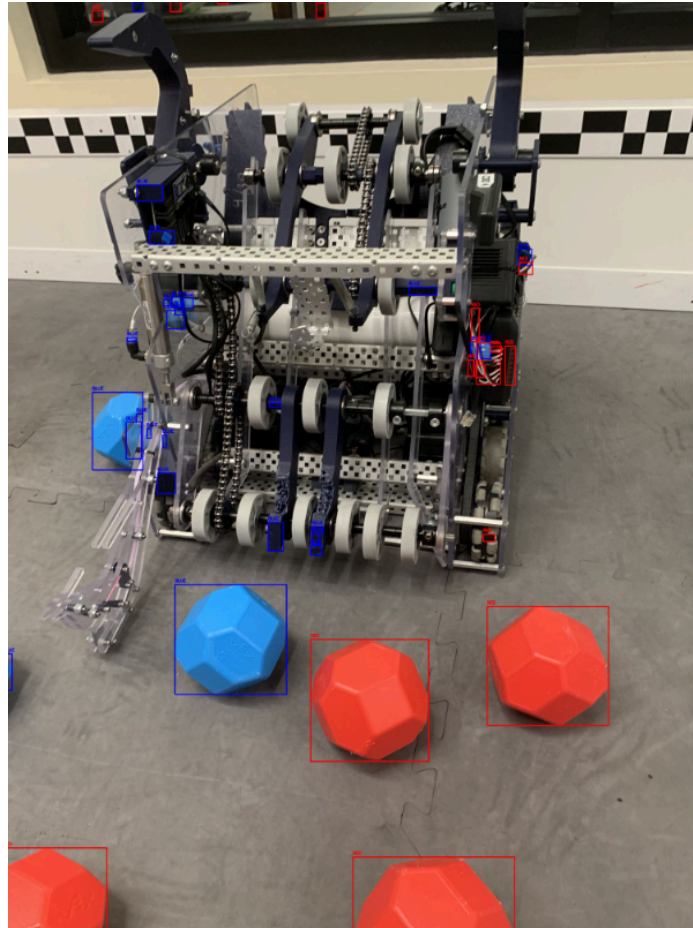


Figure 2: Color detection model

The next technique that was tested and evaluated was edge detection. The edge detection program did a really good job of finding a cluster of edges and identifying that as a robot as well as using similar techniques as the color detection to identify the field elements as well. See **Figure 3**.

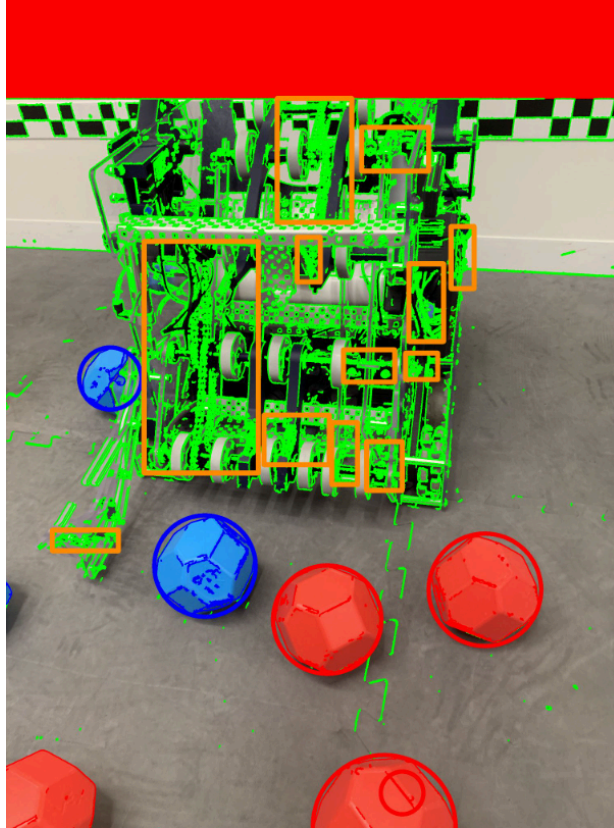


Figure 3: Edge detection model

Next writing a custom CNN architecture was attempted using tensor flow but due to compute constraints the amount of training needed for this model to perform well was not possible. See **Figure 4** for the training and validation loss chart and **Figure 5** for an attempted prediction from the model.

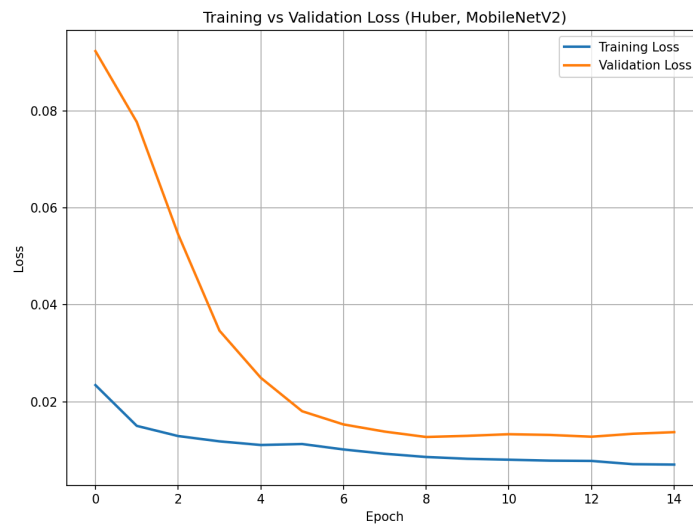


Figure 4: Training loss and validation loss graph

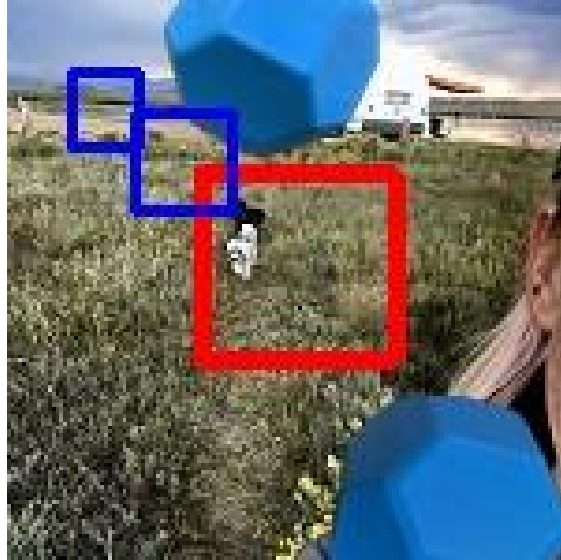


Figure 5: Attempted prediction

Since the computer needed to make the model perform accurately, pretrained YOLO weights were used to increase what the model has actually seen since data was limited. After this the model performed much better as seen in **figure 6**.

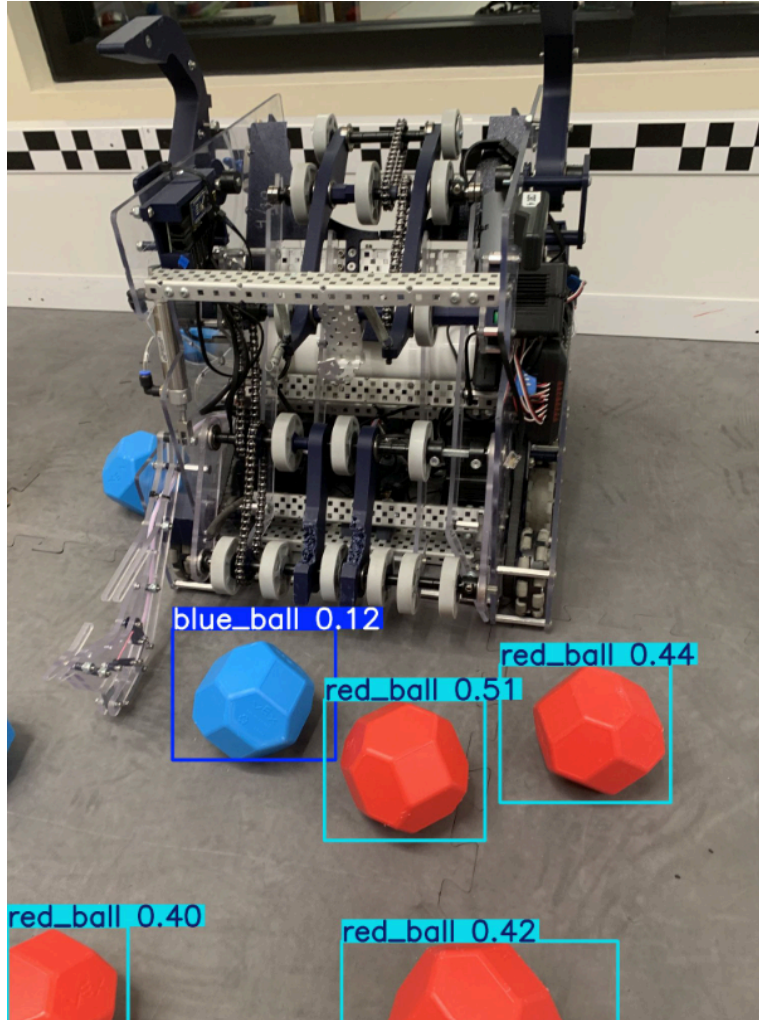


Figure 6: CNN model

After viewing the different images and observing how the overlap can be integrated it is clear that the edge noise from the field elements can be deleted by the CNN and color detection while the edge detection model can find the cluster of edges of an actual robot.

Discussion & Future Work

Key Insights

The results of this project demonstrate that combining classical computer vision algorithms with lightweight neural network-based detectors provides a balanced approach to real-time perception in VEXU robotics. Color segmentation proved highly efficient and accurate under controlled lighting, offering a strong baseline for detecting red and blue game elements. Edge-based clustering reliably identified robot structures regardless of team-to-team variability, highlighting the value of geometry-driven methods when object appearance is unpredictable. The CNN and YOLO models achieved the strongest generalization across diverse backgrounds, confirming the effectiveness of synthetic data generation for domains with limited labeled datasets. Collectively,

these findings suggest that hybrid perception pipelines offer significant advantages in environments where speed, robustness, and adaptability are equally important.

Limitations

Despite promising performance, several limitations remain. Color-based methods degrade significantly under variable lighting or when field barriers share similar hues with game elements. Edge detection, while effective for structural recognition, frequently detects irrelevant contours created by field walls or reflections, requiring further filtering or classification. The CNN models, although trained on synthetic data, still exhibit occasional failures in cases with extreme occlusion or non-standard lighting conditions not represented in the synthetic dataset. Additionally, resource constraints on embedded hardware may limit the deployment of more complex neural architectures without optimized inference engines.

Broader Impact

Improving autonomous perception in robotics competitions enhances both strategic depth and safety. Accurate robot detection helps prevent collisions and promotes fair, predictable interactions on the field. Beyond VEXU, techniques developed in this project—particularly hybrid pipelines and synthetic training data—are relevant to broader robotics domains such as warehouse automation, surveillance, and assistive robotics. Ethical considerations are minimal due to the absence of human subjects; however, broader deployment of autonomous detection systems raises general concerns about surveillance, data privacy, and unintended use cases if such methods are applied outside competitive settings.

Future Directions

Future research could explore more advanced deep-learning architectures, such as YOLOv8 or transformer-based detectors, optimized for edge devices through pruning, quantization, or TensorRT conversion. Expanding the synthetic dataset with procedurally generated robot models or domain-randomized lighting simulation may improve model robustness. Integrating temporal filtering—such as Kalman filtering or optical flow—could stabilize detections across consecutive frames, reducing flicker and improving tracking. Additional work could also incorporate depth sensing or stereo vision to estimate object distances, improving navigation and interaction planning during autonomous periods.

Lessons Learned

The project highlights the importance of balancing classical and learned techniques in constrained robotics environments. Classical algorithms offer transparency, speed, and reliable performance in well-structured settings, while neural models provide resilience to variation and ambiguity. The use of synthetic data proved essential, underscoring the value of scalable dataset generation when real-world labeling is impractical. Additionally, iterative testing in realistic conditions revealed that small implementation choices—such as HSV calibration or

edge-threshold tuning—can substantially influence system performance. These insights emphasize the importance of modular design, extensive experimentation, and adaptability in developing robust robotic vision systems.

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